

AI for All Minds in Higher Education

Executive summary

Neurodiversity in higher education is substantial, heterogeneous, and persistently undercounted. In the United States, NCES reports that 21% of undergraduates and 11% of postbaccalaureate students report having a disability, while a recent national student-access study found much higher rates when students were asked about disabling conditions more broadly, underscoring how much institutional disclosure misses. In UK higher education, reported specific learning difficulties and ADHD together rose from 4.08% of students in 2007/08 to 5.57% in 2018/19 in HESA-linked analyses. Reviews of the HE neurodiversity literature consistently find that the evidence base is still dominated by autism, ADHD, and dyslexia, with smaller literatures on dyspraxia, dysgraphia, and acquired cognitive conditions. ¹

The strongest direct evidence on AI and neurodiversity in higher education currently concerns generative AI and LLM-style chatbots. Two important recent studies found that disabled students in HE use ChatGPT and related tools for interpreting assignment briefs, summarizing reading, brainstorming, structuring writing, proofreading, and self-organization. The clearest reported benefits are reduced cognitive load, better writing scaffolds, more accessible explanations, and low-friction support for executive-function challenges. The clearest reported risks are inaccuracies, academic-integrity uncertainty, pricing inequities, inaccessible interfaces, loss of time through overuse, and dependence that may displace learning if policies and instruction are weak. ²

Outside LLMs, the evidence is more uneven. There is meaningful higher-education evidence for assistive technology overall, for note-capture and reading/writing supports, and for at least one intelligent tutoring system that improved self-regulated learning among university students with learning difficulties. There are also credible official campus case studies for AI-enabled student-support chatbots and learning analytics. But the evidence is much thinner for neurodiversity-specific recommender systems, predictive analytics evaluated by disability subgroup, and multimodal AI systems assessed specifically for neurodivergent student outcomes in HE. In other words, campuses should not assume that because a tool is “AI-enabled,” its inclusion value has been demonstrated. ³

The practical implication is not “ban AI” or “roll it out everywhere.” It is to treat AI as part of an inclusive support stack: accessible curriculum design first, accommodations and assistive technology second, AI as optional augmentation third, and human judgement always retained. The most evidence-aligned campus strategy is a phased pilot model with neurodivergent student and staff co-design, WCAG-centered procurement, explicit policies for reasonable adjustments, staff and student AI-literacy training, fairness and privacy review, and outcome measurement that looks for differential benefit or harm by condition and role. ⁴

Neurodiversity in higher education

In current higher-education research, **neurodivergence** is generally used as a neuro-affirming umbrella term for forms of cognitive difference that most commonly include ADHD, autism spectrum conditions, and dyslexia; some work also includes dyspraxia, dysgraphia, Tourette syndrome, and related profiles. A 2025 participatory priority-setting exercise in UK HE explicitly notes that neurodivergence in HE most commonly encompasses ADHD, autism, and dyslexia, while a 2024 rapid review narrowed its evidence synthesis to commonly researched conditions because the literature remains fragmented and siloed. That fragmentation matters: policy language is broader than the evidence base. ⁵

A useful analytic distinction is between **neurodevelopmental neurodivergence** and **acquired cognitive disability**. The former usually includes autism, ADHD, and specific learning difficulties; the latter includes traumatic brain injury. TBI is not always included in strict neurodiversity definitions in higher-education scholarship, but it is reasonable to include it in an “All Minds” implementation framework because TBI often creates overlapping barriers in attention, working memory, processing speed, executive function, note-taking, and fatigue. That overlap is exactly where many AI and assistive technologies operate. ⁶

What can be said confidently about prevalence is more limited than many campus presentations imply. Disability disclosure data, self-identification data, and diagnostic prevalence are not interchangeable. In the U.S., NCES places undergraduate disability prevalence at 21%, but the National Disability Center’s 2025 national student-access report found that when disability was framed around disabling conditions rather than formal campus registration, 43% of respondents in its sample indicated they had a disability. More than half of disabled students in that study reported more than one disability, which is highly relevant for AI implementation because many real users do not fit a single-condition model. ⁷

Specific-condition prevalence in HE is best treated as a patchwork rather than a clean campus census. A HESA-based longitudinal analysis found reported SpLD/ADHD prevalence around 4–6% of the UK student body across 12 years, increasing from 4.08% in 2007/08 to 5.57% in 2018/19. In Ireland, one ADHD case study reported that 5% of university students registering with disability services had ADHD, with requests for assessment and support rising annually. In the National Disability Center’s 2025 survey of disabled college students, 43% reported ADD/ADHD, 25% reported “neurodivergent,” 13% reported autism, and 5% reported learning disability. These are not the same denominator, but together they show why any campus AI strategy should expect ADHD and mixed neurodivergent profiles to be common, with autism and dyslexia also highly visible. ⁸

For TBI, the measurement problem is even greater. Postsecondary samples in the literature vary widely, and one review of post-secondary education after brain injury notes estimates for mild TBI in postsecondary samples ranging from 6% to 39%; another college study with 1,043 students specifically investigated the prevalence of TBI and associated academic consequences. The lesson is not that TBI is precisely as common as any one figure suggests, but that acquired cognitive disability is present on campus at nontrivial levels and is often poorly integrated into “neurodiversity” conversations despite direct implications for support design. ⁹

Working definitions and what prevalence data can actually support

Condition or term	Working definition for this report	What prevalence data can support
Neurodiversity / neurodivergence	A neuro-affirming umbrella concept for cognitive difference; in HE research it most often centers ADHD, autism, and dyslexia. ⁵	Broad estimates often cite 15–20% of the population as neurodivergent, but campus-specific measurement usually relies on disability disclosure and therefore undercounts. U.S. undergraduates reporting disability: 21%. ¹⁰
ADHD	Neurodevelopmental condition affecting attention regulation, impulsivity, and executive function across contexts. ¹¹	Diagnosed adult ADHD in the U.S. was estimated at 6.0% in 2023; in one Irish HEI, 5% of students registering with disability services had ADHD; among disabled students in the 2025 national access survey, 43% reported ADD/ADHD. ¹²
Autism	Neurodevelopmental profile associated with differences in social communication, sensory processing, and patterns of attention/interest. ¹³	In the disabled-student access survey, 13% reported autism; UK sources also report growth in the number of autistic students in universities, though institutional counting methods vary. ¹⁴
Dyslexia / specific learning difficulties	Specific learning profiles affecting reading, spelling, written expression, and related processing. In HE datasets, often grouped with dyspraxia and ADHD. ¹⁵	HESA-based analysis found reported SpLD/ADHD at roughly 4–6% of students over 12 years, with dyslexia historically driving much of the category. ¹⁶
Dyspraxia / dysgraphia	Motor-planning and/or written-production difficulties that can affect handwriting, note-taking, pacing, and organization. ¹⁷	Much less often reported separately in HE studies; commonly absorbed into broader SpLD groupings. ¹⁸
TBI	Acquired brain injury that can affect attention, memory, fatigue, executive function, and communication. Included here for pragmatic inclusion design, not because all neurodiversity scholars define it the same way. ¹⁹	Postsecondary estimates vary widely and are method-sensitive; available studies show TBI is relevant and often under-addressed in college support systems. ⁹

The trend line below is useful not because it resolves prevalence, but because it visualizes a robust fact: reported SpLD/ADHD presence in UK higher education rose materially over time even in administrative data, which are typically conservative. ¹⁶

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xychart-beta
  title "Reported SpLD/ADHD prevalence in UK higher education"
  x-axis [2007/08, 2011/12, 2014/15, 2018/19]
  y-axis "Percent of students" 0 --> 7
  bar [4.08, 5.15, 5.85, 5.57]
```

Current AI uses and the strength of the evidence

The most direct higher-education evidence currently available is for **LLMs and generative chatbots**. A 2025 interview study of 33 students with disabilities found that ChatGPT was used as an assistant for teaching, writing, reading, research, and self-organization. A 2025 survey of 124 students with disabilities—primarily students with ADHD, dyslexia/dyspraxia, and autism-related social/communication impairments—found that the main tools in use were chatbots, rewriting tools, and translation software. These students reported using AI to interpret assignment briefs, summarize reading, brainstorm ideas, structure assignments, rewrite phrases, and proofread. The three affordances highlighted most strongly in the 2026 follow-on study were explainability, expressibility, and plannability. ²⁰

For **adaptive learning and intelligent tutoring systems**, the direct HE evidence is narrower but not trivial. In a university experiment with 119 students with and without learning difficulties, an intelligent tutoring system improved the use of self-regulated learning strategies, and students with learning difficulties increased their deep self-regulated learning strategies when prompted by pedagogical agents. This is more specific and more rigorous than generic marketing claims about “personalization,” but it also came from one experimental setting rather than a mature body of campus replication studies. ²¹

For **speech-to-text, text-to-speech, note-capture, and traditional assistive technology**, the evidence base is older and broader than current GenAI evidence. A 2021 systematic review of assistive technology use in higher education synthesized 26 papers and identified four recurring themes: assistive technology as an enabler of academic engagement, barriers to effective use that can hinder engagement, transformative psychosocial effects, and improved participation. Earlier postsecondary studies found that students with dyslexia were generally positive about assistive technology provision but that training quality mattered, and small postsecondary intervention studies found that OCR plus speech-synthesis supports could improve reading comprehension for students with learning disabilities. ²²

For **smartpens and note-capture tools**, the evidence is smaller but practically relevant. Research and case studies involving college students with dyslexia or other disabilities suggest that smartpens can reduce note-taking load, support lecture review, and improve confidence. A Berkeley disability-services pilot reported providing smartpens to 40 students with a range of physical and learning disabilities, and more recent conference work suggests smartpen-plus-structured-note strategies can improve comprehension and achievement motivation among university students with disabilities. ²³

For **virtual assistants and chatbots in disability services**, the Open University case is the strongest official HE example. The “Taylor” digital assistant was developed through participatory research with disabled students, designed as an alternative to forms for disability disclosure and support conversations, and later integrated into university systems after trials. Official OU and Jisc material describes its value in

reducing administrative burden, enabling text or speech interaction, and helping students understand available support while allowing staff to use their time more effectively. ²⁴

For **predictive analytics and learning analytics**, institutions are using AI-like or data-driven systems now, but neurodiversity-specific evidence is weak. Jisc documents current higher-education use for retention, attendance, case management, and wellbeing support, including a Newcastle University example where engagement and attendance data help wellbeing staff identify students who may be struggling and support more informed conversations. At the same time, Jisc’s own accessibility guidance warns that disabled students can be misclassified as “at risk” when alternative learning strategies or accommodation patterns are not reflected in the model. ²⁵

For **recommender systems and multimodal AI**, the promise is real, but direct HE neurodiversity evidence is sparse. Some work explores recommendation models for matching dyslexic learners to supportive tools, and sector reports emphasize multimodal AI for image description, audio/text flexibility, and accessible scheduling or guidance. But compared with LLM use, the evidence for measurable higher-ed outcomes among neurodivergent students remains thin. Campuses should therefore classify these tools as “promising but not yet well-validated” rather than “proven.” ²⁶

Tool comparison

Tool type	What “use now” looks like in higher education	Representative evidence	Benefits supported by evidence	Main harms or limits	Recommended campus action
LLMs and chatbots	Assignment interpretation, summarization, idea generation, structure, proofreading, study planning, basic Q&A. ²⁷	33 interviews with students with disabilities; 124-student survey; 2026 affordance study. ²⁸	Better access to reading and writing tasks; scaffolds for executive function and cognitive load. ²⁹	Inaccuracy, integrity ambiguity, cost barriers, accessible-design gaps, overuse and time sink. ³⁰	Offer campus-approved tools, explicit assessment rules, training in prompting, verification, and attribution. ³¹
Adaptive learning and ITS	Prompted self-regulation, stepwise scaffolds, personalized strategy prompts. ²¹	University experiment with 119 students. ²¹	Improved self-regulated learning, especially with prompts for students with learning difficulties. ²¹	Limited replication in ND-specific HE contexts; personalization claims often exceed evidence. ³²	Pilot only in clearly bounded courses; evaluate subgroup effects before scale. ³²

Tool type	What “use now” looks like in higher education	Representative evidence	Benefits supported by evidence	Main harms or limits	Recommended campus action
Speech-to-text, text-to-speech, captioning	Reading support, writing dictation, lecture review, accessibility overlays, captions/transcripts. ³³	Systematic review of AT in HE; postsecondary dyslexia and LD studies; higher-ed captioning discussions. ³⁴	Strong practical value for reading, writing, and lecture access; can reduce decoding and note-taking burden. ³⁵	Auto-captions may be inaccurate; tool benefit depends on training, compatibility, and workflow. ³⁶	Treat as baseline access infrastructure, not optional add-ons; quality-check captions. ³⁷
Note-capture and smartpens	Audio-synced notes, review support, lecture replay, reduced note-taking load. ³⁸	Dyslexia-focused studies and institutional pilots. ²³	Better lecture review, anxiety reduction, improved confidence and motivation. ³⁹	Works best when course recording policies and privacy expectations are clear. ⁴⁰	Include in assistive-tech menus and accommodation workflows. ⁴¹
Virtual assistants for disability services	Disclosure, triage, guidance about supports, conversational replacement for forms. ⁴²	Open University participatory case study and official roll-out reports. ⁴³	Lower administrative burden, more accessible entry point, better student preparation for human support. ⁴⁴	Confidentiality, scope creep, and risk of over-automating sensitive processes. ⁴⁵	Use as front door, not final decision-maker; preserve human escalation. ⁴⁶
Predictive analytics and learning analytics	Early-alert flags, attendance and VLE monitoring, case management, wellbeing triage. ⁴⁷	Official Jisc platform documentation; Newcastle wellbeing case. ⁴⁷	Earlier identification of disengagement and more coordinated support. ⁴⁷	Can misclassify disabled students; confidentiality and inference risks; automated nudges may be poorly targeted. ⁴⁸	Require fairness review, human mediation, and disability-aware thresholds before intervention. ⁴⁹

Tool type	What “use now” looks like in higher education	Representative evidence	Benefits supported by evidence	Main harms or limits	Recommended campus action
Recommenders and multimodal AI	Tool suggestions, image description, text/speech interaction, accessible scheduling and support pathways. ²⁶	Sparse and emerging evidence. ⁵⁰	Potentially strong for fit-to-learner variation and reducing text-only burden. ⁵¹	Weak HE outcome evidence; can reproduce bias if training data ignore disability variation. ⁵²	Pilot as optional enhancement, not as core accommodation replacement. ⁵³

Differential benefits, harms, and role-based impacts

The most important analytical point is that AI does **not** affect “neurodivergent students” as a single group. The most direct recent studies show distinct patterns by condition. Students with ADHD, for example, often report that LLMs help with idea organization, writer’s block, and planning, but the same tools can also become a productivity trap: some interviewees reported wasting time on the platform, and several students with ADHD noted that the real challenge is not making a plan but sticking to it. That distinction is crucial for campus practice. A tool that is strong at planning but weak at follow-through is not an executive-function solution on its own. ⁵⁴

For autistic students, the evidence suggests value in predictable, low-judgment, asynchronous support. Interviewees described using ChatGPT to condense long, hard-to-focus-on readings and to ask questions they felt uncomfortable asking people. That can lower social exposure costs and make support feel more available. But autism also appears in the risk literature: AI systems that rely on facial-recognition or emotion-inference logics can be poorly trained on autistic presentations, and disability advocates warn that accessibility claims often overpromise what AI can infer. ⁵⁵

For dyslexia and broader specific learning difficulties, the evidence is strongest around reading and writing supports. Students report value in proofreading, explanation of mistakes, summarization, and help getting started; older but still important postsecondary literature also shows that TTS, OCR, and related assistive technologies can improve access and academic engagement when students are trained to use them. At the same time, text-heavy AI interfaces can themselves become barriers, and “feature creep” in tools like grammar checkers or translators can turn a legitimate accommodation into a policy problem if assessment rules are vague. ⁵⁶

For dyspraxia, dysgraphia, and writing-production difficulties, the evidence is usually folded into broader SpLD categories rather than studied independently. That is itself a research gap. Still, the practical case for speech-to-text, rewriting support, multimodal input, and reduced transcription burden is strong enough to justify cautious implementation as long as assessment validity is protected. ⁵⁷

For TBI, the literature is markedly thinner on AI specifically, so any claim must be more modest. What the postsecondary TBI literature does show is that students frequently face challenges in memory, attention, executive function, fatigue, and study strategy use, and that supports such as recording, note provision, extra time, structured instructions, and assistive technology matter. From an implementation standpoint, that means transcription, note-capture, summarization, and stepwise AI support are **plausibly helpful**, but institutions should be honest that this is currently more of an inference from support-needs research than a well-established AI-efficacy result in HE. ⁵⁸

The harms are just as differentiated. Predictive analytics can unintentionally single out disabled students as “at risk” when their attendance patterns reflect accommodation rather than disengagement. Students using paid LLM features may gain support others cannot afford. Auto-captioning can falsely signal accessibility when accuracy is too low to be equivalent access. And anti-AI responses—such as shifting back to handwritten, in-class, tightly timed assessments or relying on unreliable AI detection—can disproportionately burden students who depend on assistive technology and flexible modalities. ⁵⁹

Differential impacts by condition

Condition	Where benefits appear strongest	Risks that appear especially salient	Confidence
ADHD	Idea structuring, overcoming mental blocks, study planning, time-management scaffolds. ⁶⁰	Time loss, rabbit-hole use, simplified information that may weaken focus, tools that help plan but do not help enact plans. ⁶¹	Moderate for LLMs; weaker for other AI types. ⁶²
Autism	Reading summarization, predictable/asynchronous support, low-judgment Q&A, communication preparation. ⁶³	Misrepresentation in AI training data; emotion/ facial-recognition harms; social/identity concerns in disclosure contexts. ⁶⁴	Moderate for LLM use; low for other AI modalities in HE. ⁶⁵
Dyslexia / SpLD	Proofreading, explanation of errors, text simplification, TTS/ OCR, note-capture. ⁶⁶	Text-heavy interfaces, integrity ambiguity in “assistive” tools with generative features, uneven training and fit. ⁶⁷	Moderate-to-strong for assistive tech broadly; moderate for LLMs. ⁶⁸
Dyspraxia / dysgraphia	Writing production support, multimodal input, reduced manual transcription. ⁶⁹	Independent evidence is sparse; risk of one-size-fits-all SpLD policies. ¹⁸	Low-to-moderate.

Condition	Where benefits appear strongest	Risks that appear especially salient	Confidence
TBI	Transcription, note-capture, structured prompts, memory/executive-function supports. This is presently an inference from support-needs and assistive-tech literature more than direct AI trials. ⁷⁰	Fatigue, variable day-to-day capacity, privacy concerns, need for human accommodation decisions. ⁷¹	Low for AI-specific HE evidence.

Differential impacts by staff role

Staff role	Where AI can help	Where it can harm	Recommended safeguard
Instructors	Plain-language explanations, study guides, multiple representations, assessment redesign, responsive support to writing and reading barriers. ⁷²	Vague or punitive policies, inaccessible assessment redesign, over-reliance on AI detectors, failure to distinguish accommodation from outsourcing. ⁷³	Publish task-level allowed/prohibited uses and align policies to learning outcomes, not blanket bans. ⁷⁴
Disability services	Intake, triage, pre-advising information, conversational disclosure, scheduling and form support. ⁷⁵	Confidentiality leaks, over-automation of complex judgment, excluding students who prefer human interaction. ⁷⁶	Keep AI at the front door, not the final decision point; require human escalation. ⁴⁶
Counselors and wellbeing teams	Earlier visibility into disengagement trends; better-informed outreach conversations. ⁴⁷	False inference of distress or disability; insensitive nudges; excessive surveillance burdens. ⁴⁸	Use analytics as conversation starter, never as diagnosis; review intervention language for psychological impact. ⁴⁸
IT, edtech, and procurement teams	Standardized accessibility review, campus licensing, integrations, protected-mode deployments. ⁷⁷	Buying inaccessible tools or tools that silently add generative features that change assessment risk. ⁷⁸	Require accessibility evidence, data controls, and change-management clauses in procurement. ⁷⁹

Implementation framework for inclusive AI

The best available evidence does **not** support a “technology first” implementation model. It supports an **inclusive-design first** model in which AI is layered onto accessible teaching, accommodations, and student services. HE neurodiversity reviews emphasize UDL, explicit instruction, strategy instruction, mentoring/coaching, and institutional neuroinclusion more consistently than they support any single AI product. Neurodivergent scholars and advocates also argue for a systems approach that includes training, transition supports, accessible communication, and neurodivergent leadership in campus decision-making. ⁸⁰

From the AI literature specifically, several principles recur across strong sources. First, **co-design with disabled and neurodivergent users** is non-negotiable; both sector reports and peer-reviewed studies warn that disabled users are rarely adequately represented in AI development. Second, **reasonable adjustments must survive AI policy**; institutional guidance from the Open University explicitly allows generative AI as a reasonable adjustment when it does not defeat the purpose of the assessment, while Sheffield’s guidance clearly distinguishes assistive use from misconduct while warning about feature creep. Third, **human mediation remains essential** in disability services and wellbeing use cases. Fourth, **privacy and data minimization** are central: official guidance warns students not to upload confidential or personal data into public tools. ⁸¹

A practical campus design stack should therefore begin with the following questions before any pilot launches: Is the underlying course material accessible without AI? Are there clear task-level rules for AI use? Will the tool work with screen readers, keyboard navigation, captions, dictation, and alternative formats? Can students access an institutionally licensed version rather than paying individually? Can the campus disable or document new generative features when they change the integrity implications of a tool? Can disability services override defaults? Those are procurement and governance questions as much as educational ones. ⁸²

Recommended campus actions

Domain	Evidence-based action	Why it is well supported
Design	Use UDL-style multiple means of representation and expression before adding AI. ⁸⁰	Neurodiversity reviews consistently support explicit instruction, clarity, and structural flexibility more strongly than any single AI tool. ⁸³
Accommodations	Permit AI as a reasonable adjustment when it supports access without invalidating the assessed skill. ⁸⁴	This matches official institutional policy and avoids penalizing students for legitimate access supports. ⁸⁵
Training	Train students and staff in AI literacy: prompting, verification, attribution, privacy, and tool limits. ⁸⁶	Disabled-student studies repeatedly call for institutional training and policy clarity. ⁸⁷
Evaluation	Measure differential outcomes by disability group and role, not just overall satisfaction. ⁸⁸	The literature is fragmented and condition-specific; aggregate gains can hide subgroup harms. ⁸⁸

Domain	Evidence-based action	Why it is well supported
Procurement	Require WCAG-aligned accessibility evidence, assistive-tech compatibility, privacy controls, roadmaps for feature changes, and equitable licensing. ⁸⁹	Accessibility and equitable access should be built into purchasing, not remediated after deployment. ⁷⁹
Ethics and policy	Ban AI detection from being the primary basis for discipline; require human review and disability-aware interpretation of evidence. ⁹⁰	Anti-AI responses can disproportionately harm disabled students, and detection reliability is poor. ⁹¹
Student services	Use AI chatbots for routing, triage, and scheduling, but preserve human follow-up and consent-sensitive data handling. ⁹²	This is where the official case-study evidence is strongest. ⁹³
Wellbeing	Use analytics only as prompts for context-rich outreach, not as proxies for disability or diagnosis. ⁹⁴	Accessibility guidance explicitly warns against false inferences and unfair singling out. ⁴⁸

Evaluation metrics that matter

Metric family	Practical examples	What to watch for
Access	Tool uptake by disability subgroup; failure rates with screen readers, captions, dictation, keyboard navigation; time-to-access supports. ⁹⁵	High overall use with low use among autistic, dyslexic, or TBI students can indicate hidden barriers.
Learning	Assignment completion, reading comprehension, writing quality, self-regulated learning, time-on-task, revision quality. ⁹⁶	Improvements should be compared by condition and by course type, not only in aggregate.
Equity	Differential benefit/harm by disability condition, cost burden, rates of accommodation conflict, false “at risk” flags. ⁹⁷	Paid-feature gaps and misclassification are common sources of inequity.
Experience	Self-efficacy, trust, belonging, cognitive load, emotional labor of disclosure, perceived autonomy. ⁹⁸	Students may like a tool while still experiencing hidden stress or surveillance concerns.
Operations	Disability-office response times, counselor triage efficiency, staff training completion, privacy incidents, complaint rates. ⁹⁹	Efficiency gains are only acceptable if access quality remains high.

The adoption pathway below reflects the strongest cross-source consensus: accessibility baseline first, co-design second, tightly scoped pilots third, and scale only after subgroup evaluation. ¹⁰⁰

flowchart LR

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A[Month 0-1\nSet governance\nInclude ND students, faculty,\ndisability services, counseling, IT] --> B[Month 1-2\nAudit baseline accessibility\ncourses, assessments, workflows, data flows]
B --> C[Month 2-3\nChoose low-risk use cases\nreading/writing support, \nintake chatbot, scheduling]
C --> D[Month 3-4\nProcure or sandbox tools\nprivacy review, accessibility testing,\nlicensing review]
D --> E[Month 4-6\nRun pilot\ntraining for students and staff\nhuman support kept in loop]
E --> F{Evaluate by subgroup\nand staff role}
F -- Problems found --> G[Revise or stop\nadjust policy, prompts, \nthresholds, or tool choice]
F -- Benefits outweigh harms --> H[Month 6-9\nFormalize policy\nreasonable adjustments,\nintegrity guidance,\nprocurement clauses]
H --> I[Month 9-12\nScale carefully\nongoing audits,\nfeedback loops, \nannual review]
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Research gaps and prioritized questions

The literature is moving fast, but it is still methodologically immature for many decisions campuses want to make. Recent reviews say the evidence remains US-centric, based on small samples, and focused more on learning processes than on successful completion, wellbeing, or long-term outcomes. A 2025 participatory priority-setting exercise further shows that the field itself sees staff knowledge, assessment, support, outcomes, and accessibility as the top unresolved domains. ¹⁰¹

The most important gaps are these:

1. **Condition-specific effectiveness.** Which AI supports genuinely help ADHD, autism, dyslexia, dyspraxia, and TBI in different academic tasks, and which simply feel helpful? The current literature often collapses these groups together. ¹⁰²
2. **Learning outcomes beyond self-report.** We need more trials that measure course performance, persistence, skill acquisition, and overreliance, not just satisfaction or perceived usefulness. ¹⁰³
3. **Fairness in predictive systems.** There is very little published HE research showing whether early-alert systems systematically overflag or under-serve neurodivergent or TBI-affected students. This is a major gap given current sector deployment. ¹⁰⁴
4. **Reasonable-adjustment boundaries.** Institutions need stronger empirical guidance on when AI use still preserves the intended learning outcome and when it substitutes for the skill being assessed. Current policy examples are sensible, but still mostly principle-based. ¹⁰⁵

5. **Staff outcomes, not only student outcomes.** Disability-services professionals, counselors, instructors, and TAs are deeply affected by AI implementation, yet the evidence on workload, judgement quality, demoralization, and role change is still limited. ¹⁰⁶
6. **TBI and acquired cognitive disability.** Compared with autism, ADHD, and dyslexia, the HE AI evidence for TBI is strikingly thin despite clear overlap in support needs. ¹⁰⁷
7. **Procurement and feature drift.** We know very little about how changes to commercial tools alter real accessibility, integrity risk, and accommodation fit over time. The “feature creep” problem is already visible in institutional guidance, but under-studied empirically. ⁷⁸

A useful near-term research agenda would therefore prioritize: subgroup-sensitive trials of writing and reading supports; fairness audits of learning analytics; longitudinal studies of learning and dependency; participatory design with neurodivergent students and staff; and implementation studies that compare “accessible baseline only” versus “accessible baseline plus AI augmentation.” ¹⁰⁸

Slide-ready presentation materials

Slide: Why this topic matters

- Neurodiversity is common on campus, but disclosure data undercount it.
- Current HE evidence centers ADHD, autism, and dyslexia; TBI is relevant but less studied.
- AI is already being used by disabled students and by universities—even where policy lags behind practice.

Speaker notes: Open by reframing the issue from “special accommodation” to “mainstream learner variability.” In the U.S., 21% of undergraduates report disability, but broader self-identification methods suggest the real share of students dealing with disability-related barriers is higher. Emphasize that campuses are not deciding whether AI exists; they are deciding whether it will be implemented inclusively or haphazardly. ¹⁰⁹

Slide: What neurodiversity means in higher education

- In HE research, neurodivergence usually means ADHD, autism, and dyslexia.
- Dyspraxia and dysgraphia are discussed less often.
- TBI should be included in “All Minds” implementation because support needs overlap, even if definitions differ.

Speaker notes: Clarify that “neurodiversity” is partly a social concept and partly an operational category used in research. Then make the pragmatic move: if a tool is meant to reduce barriers in attention, memory, writing, reading, or cognitive load, TBI belongs in the implementation conversation whether or not every scholar uses the same label. ¹¹⁰

Slide: Where AI is helping now

- LLMs/chatbots: summarizing, structuring writing, interpreting briefs, proofreading.
- Traditional assistive tech: speech-to-text, text-to-speech, note-capture, captioning.
- Student-support AI: disclosure chatbots, scheduling, guidance, early-alert dashboards.

Speaker notes: The strongest direct evidence is for LLMs in reading and writing workflows. A second

cluster is established assistive technology, which predates the current GenAI wave. A third cluster is administrative and student-support AI, where official case studies are often stronger than formal trials. 111

Slide: What the best studies show

- Disabled students use AI for real academic barriers, not only convenience.
- Benefits are clearest in reading, writing, and executive-function support.
- Evidence is strongest for LLMs and assistive tech, weaker for predictive analytics and recommender systems.

Speaker notes: Mention the two anchor studies: 33 interviews and 124 survey responses from disabled students in HE. Then add the ITS experiment and the systematic review of assistive technology. This helps the audience distinguish between direct evidence, adjacent evidence, and hype. 112

Slide: Differential impacts across conditions

- ADHD: strong for idea organization and planning, weaker for follow-through.
- Autism: strong for predictable, low-judgment support and summarization.
- Dyslexia/SpLD: strong for reading/writing supports and proofreading.
- TBI: plausible gains in note-taking and structure, but AI-specific evidence is thin.

Speaker notes: Stress that “one neurodivergent policy” is too blunt. A useful way to say this aloud is: the same prompt box does different things for different learners. Planning support helps some ADHD students a great deal, but can also become a distraction. Dyslexic students may benefit from simplification and proofreading; autistic students may particularly value asynchronous nonjudgmental support. 113

Slide: The main risks

- Inaccurate outputs and academic-integrity confusion.
- Paid features can widen inequity.
- Predictive systems can misclassify disabled students.
- Anti-AI backlash can unintentionally block real accommodations.

Speaker notes: Do not frame risk only as “students cheating.” The better framing is “misfit between tool, policy, and learner.” A caption that is inaccurate is an accessibility failure. An early-alert flag that misreads accommodation patterns is a fairness failure. A blanket ban that forces handwritten or AI-detector-policed assessment can also be a disability failure. 114

Slide: What an inclusive implementation looks like

- Accessibility baseline first.
- Co-design with neurodivergent students and staff.
- Allow AI as reasonable adjustment when learning outcomes are preserved.
- Train everyone in verification, privacy, and disclosure expectations.
- Keep humans in the loop.

Speaker notes: This is the “AI for All Minds” core message. Accessibility does not begin with AI. It begins with accessible courses, clear instructions, and a service model that does not force students

through unnecessary friction. Then AI becomes a flexible layer of support rather than a fragile workaround. 115

Slide: How to pilot it

- Start with low-risk use cases.
- Measure outcomes by subgroup and staff role.
- Stop, revise, or scale based on evidence.
- Build policy and procurement after, not before, a real user pilot.

Speaker notes: Recommend starting with reading/writing support, scheduling, and triage rather than predictive surveillance or high-stakes assessment uses. The most important evaluation question is not “did students like it?” but “who benefited, who was left behind, and did any subgroup experience new barriers?” 116

Slide: Final takeaways

- AI can widen inclusion, but only if accessibility and policy are designed together.
- The best-supported uses right now are scaffolding, not substitution.
- Campuses should treat AI as optional augmentation inside an inclusive ecosystem, not as a shortcut around disability support.

Speaker notes: End on a balanced note. The evidence does not support techno-solutionism, but it also does not support panic. The report’s practical conclusion is simple: use AI to reduce friction and cognitive load, not to replace pedagogy, disability services, or human judgement. 117

1 7 109 <https://nces.ed.gov/fastfacts/display.asp?id=60>
<https://nces.ed.gov/fastfacts/display.asp?id=60>

2 20 28 29 54 55 56 60 61 62 63 65 66 111 112 113 117 <https://link.springer.com/article/10.1007/s10639-024-13134-8>
<https://link.springer.com/article/10.1007/s10639-024-13134-8>

3 21 32 96 103 <https://www.mdpi.com/2071-1050/12/21/9184>
<https://www.mdpi.com/2071-1050/12/21/9184>

4 100 115 <https://teaching.uoregon.edu/sites/default/files/2022-02/dwyer-et-al-2022.neuroinclusive-higher-ed.pdf>
<https://teaching.uoregon.edu/sites/default/files/2022-02/dwyer-et-al-2022.neuroinclusive-higher-ed.pdf>

5 10 13 15 108 110 <https://www.mdpi.com/2227-7102/15/7/839>
<https://www.mdpi.com/2227-7102/15/7/839>

6 80 83 88 101 102 <https://journals.sagepub.com/doi/10.1177/27546330241291769>
<https://journals.sagepub.com/doi/10.1177/27546330241291769>

8 16 <https://pmc.ncbi.nlm.nih.gov/articles/PMC12992651/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12992651/>

9 <https://pubs.asha.org/doi/10.1044/sbi5.3.19>
<https://pubs.asha.org/doi/10.1044/sbi5.3.19>

- 11 <https://www.cdc.gov/adhd/articles/adhd-across-the-lifetime.html>
<https://www.cdc.gov/adhd/articles/adhd-across-the-lifetime.html>
- 12 <https://www.cdc.gov/mmwr/volumes/73/wr/mm7340a1.htm>
<https://www.cdc.gov/mmwr/volumes/73/wr/mm7340a1.htm>
- 14 <https://nationaldisabilitycenter.org/wp-content/uploads/2025/02/Student-Access-Report-2025-Accessible.pdf>
<https://nationaldisabilitycenter.org/wp-content/uploads/2025/02/Student-Access-Report-2025-Accessible.pdf>
- 17 27 30 51 52 57 64 67 69 86 87 114 <https://eprints.whiterose.ac.uk/id/eprint/224322/3/use%20of%20generative%20ai%20by%20disabled%20students%20accepted%20version%20.pdf>
<https://eprints.whiterose.ac.uk/id/eprint/224322/3/use%20of%20generative%20ai%20by%20disabled%20students%20accepted%20version%20.pdf>
- 18 https://neurodiversity-engineering.media.uconn.edu/wp-content/uploads/sites/3154/2022/02/Clouder2020_Article_NeurodiversityInHigherEducatio.pdf
https://neurodiversity-engineering.media.uconn.edu/wp-content/uploads/sites/3154/2022/02/Clouder2020_Article_NeurodiversityInHigherEducatio.pdf
- 19 58 70 107 <https://files.eric.ed.gov/fulltext/EJ966126.pdf>
<https://files.eric.ed.gov/fulltext/EJ966126.pdf>
- 22 33 34 35 41 68 https://mural.maynoothuniversity.ie/13596/1/DD_the%20impact.pdf
https://mural.maynoothuniversity.ie/13596/1/DD_the%20impact.pdf
- 23 38 40 <https://www.ischool.berkeley.edu/news/2011/computerized-pen-helps-students-disabilities-unexpected-ways>
<https://www.ischool.berkeley.edu/news/2011/computerized-pen-helps-students-disabilities-unexpected-ways>
- 24 43 46 92 93 98 <https://www.mdpi.com/2227-7102/11/10/587>
<https://www.mdpi.com/2227-7102/11/10/587>
- 25 47 104 <https://www.jisc.ac.uk/learning-analytics>
<https://www.jisc.ac.uk/learning-analytics>
- 26 50 <https://arxiv.org/html/2403.14710v1>
<https://arxiv.org/html/2403.14710v1>
- 31 78 <https://sheffield.ac.uk/study-skills/digital/generative-ai/assistive-technology>
<https://sheffield.ac.uk/study-skills/digital/generative-ai/assistive-technology>
- 36 37 <https://nationaldeafcenter.org/resource-items/automatic-captions/>
<https://nationaldeafcenter.org/resource-items/automatic-captions/>
- 39 <https://infonomics-society.org/wp-content/uploads/Assistive-Technology-and-Note-Taking-Strategies-to-Foster-Achievement-Motivation.pdf>
<https://infonomics-society.org/wp-content/uploads/Assistive-Technology-and-Note-Taking-Strategies-to-Foster-Achievement-Motivation.pdf>
- 42 44 75 <https://nationalcentreforai.jiscinvolve.org/wp/2021/09/27/how-digital-assistants-are-promoting-enhanced-accessibility-at-the-open-university/>
<https://nationalcentreforai.jiscinvolve.org/wp/2021/09/27/how-digital-assistants-are-promoting-enhanced-accessibility-at-the-open-university/>

45 48 49 59 76 97 116 <https://analytics.jiscinvolve.org/wp/2016/12/14/accessibility-considerations-for-learning-analytics/>

<https://analytics.jiscinvolve.org/wp/2016/12/14/accessibility-considerations-for-learning-analytics/>

53 77 79 82 89 90 91 95 <https://www.everylearnereverywhere.org/wp-content/uploads/Where-AI-Meets-Accessibility-Final.pdf>

<https://www.everylearnereverywhere.org/wp-content/uploads/Where-AI-Meets-Accessibility-Final.pdf>

71 <https://files.eric.ed.gov/fulltext/EJ1133819.pdf>

<https://files.eric.ed.gov/fulltext/EJ1133819.pdf>

72 73 106 [https://eric.ed.gov/?](https://eric.ed.gov/?id=EJ1475099&q=source%3A%22New+Directions+for+Teaching+and+Learning%22)

[id=EJ1475099&q=source%3A%22New+Directions+for+Teaching+and+Learning%22](https://eric.ed.gov/?id=EJ1475099&q=source%3A%22New+Directions+for+Teaching+and+Learning%22)

<https://eric.ed.gov/?id=EJ1475099&q=source%3A%22New+Directions+for+Teaching+and+Learning%22>

74 84 85 105 <https://about.open.ac.uk/policies-and-reports/policies-and-statements/generative-ai-learning-teaching-and-assessment-ou-0>

<https://about.open.ac.uk/policies-and-reports/policies-and-statements/generative-ai-learning-teaching-and-assessment-ou-0>

81 <https://er.educause.edu/articles/2024/9/the-impact-of-ai-in-advancing-accessibility-for-learners-with-disabilities>

<https://er.educause.edu/articles/2024/9/the-impact-of-ai-in-advancing-accessibility-for-learners-with-disabilities>

94 <https://www.jisc.ac.uk/podcasts/beyond-the-technology-using-jisc-learning-analytics-to-support-wellbeing>

<https://www.jisc.ac.uk/podcasts/beyond-the-technology-using-jisc-learning-analytics-to-support-wellbeing>

99 <https://iet.open.ac.uk/research/ou-launches-new-chatbot-for-students-to-disclose-disabilities-with-ease>

<https://iet.open.ac.uk/research/ou-launches-new-chatbot-for-students-to-disclose-disabilities-with-ease>